

PySpark Installation Guide for Windows

Follow the steps below to install Spark on Windows and configure Python for PySpark.

Installing Prerequisites:

1. Java

PySpark requires Java 1.8.x or above

Check if Java is already available by opening the command prompt and giving the command as shown here.

```
C:\>java -version
java version "1.8.0_471"
Java(TM) SE Runtime Environment (build 1.8.0_471-b09)
Java HotSpot(TM) 64-Bit Server VM (build 25.471-b09, mixed mode)
```

If not available, then download JDK that is appropriate to your system from official Oracle website from the link below and install it by double-clicking to run the downloaded file.

<https://www.oracle.com/in/java/technologies/javase/javase8u211-later-archive-downloads.html>

OR

<https://www.oracle.com/in/java/technologies/javase/jdk11-archive-downloads.html>

After installing check as shown above and make sure the correct version is installed and available. Also, make a note of the folder where it is installed as we will need the path later.

2. Python

Python 3.8.x to 3.11.x is required to run PySpark. Note that Python versions above 3.11.x are not compatible with current version of PySpark as of now.

Check if a compatible version of Python is already available, from the command prompt as shown here.

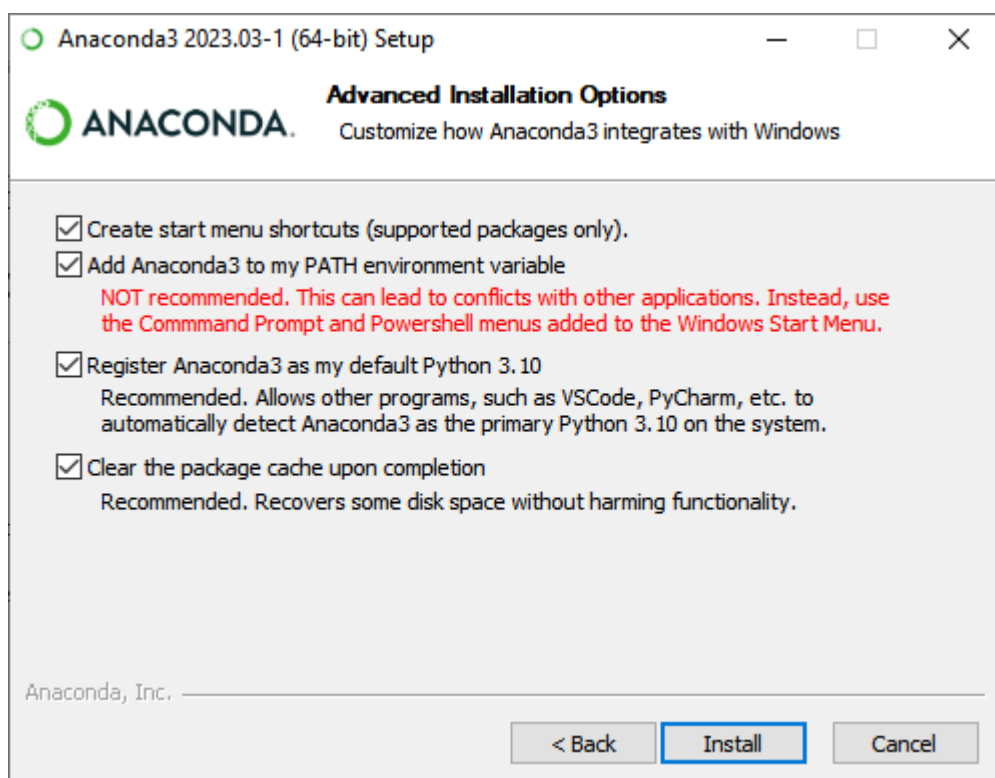
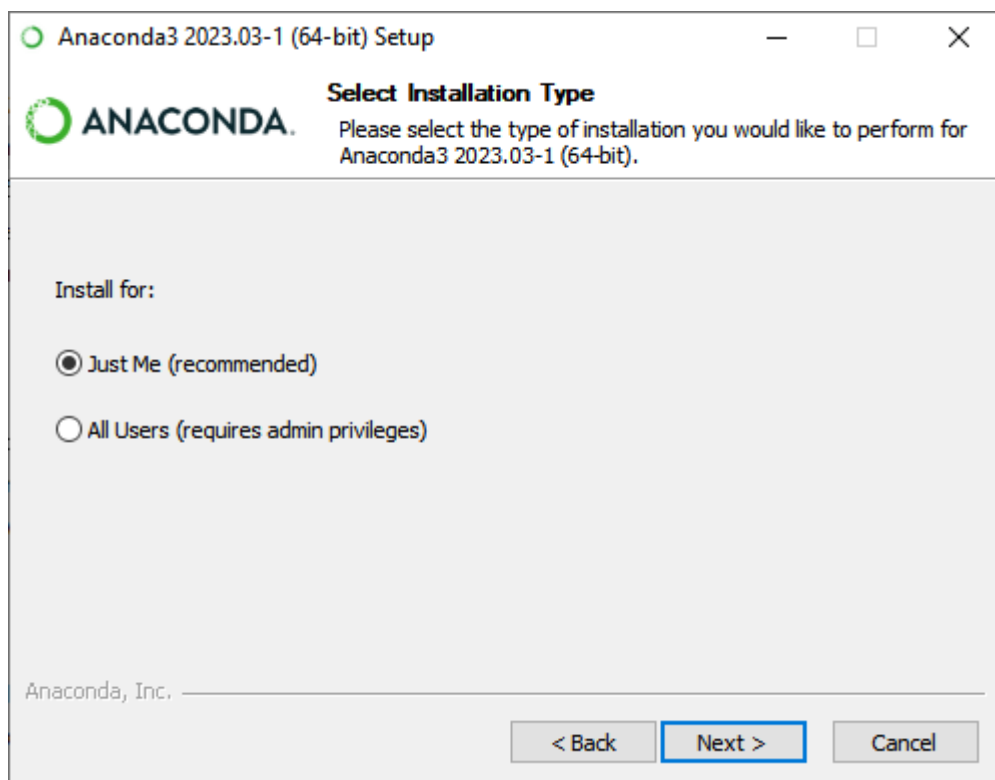
```
C:\>python --version
Python 3.10.10
```

If not available and if Anaconda is the preferred environment then you can download it from the link below and install it.

https://repo.anaconda.com/archive/Anaconda3-2025.12-2-Windows-x86_64.exe

If Anaconda is already installed then make sure that the following are added to the path.

```
<YourDrive>:\<YourDir>\anaconda3;
<YourDrive>:\<YourDir>\anaconda3\Scripts;
<YourDrive>:\<YourDir>\anaconda3\Library\bin;
<YourDrive>:\<YourDir>\anaconda3\Library\usr\bin;
<YourDrive>:\<YourDir>\anaconda3\Library\mingw-w64\bin
```



Installing Spark:

Download Spark from Apache Spark web site:

<https://spark.apache.org/downloads.html>

In the dropdown *Choose a Spark release*: select a release 3.5.X

In the dropdown *Choose package type*: Pre-built for Apache Hadoop 3.3 and later

Download Apache Spark™

1. Choose a Spark release: ▼
2. Choose a package type: ▼
3. Download Spark: [spark-3.5.8-bin-hadoop3.tgz](#)
4. Verify this release using the 3.5.8 [signatures](#), [checksums](#) and [project release KEYS](#) by following these [procedures](#).

Copy the above file into a folder `C:\Spark` Make sure not to use spaces in the folder names.

Extract the files into a sub-folder say `spark-3.5.X-bin-hadoop3` using a utility like 7-Zip or WinRAR that is available on your system.

Note that if you use 7-Zip, from the above `.tgz` file it first extracts `.tar` file. You need to use 7-Zip on the `.tar` file one more time and extract the files and folders from it.

Make a note of the folder where it is installed as we will need the path later.

Adding Hadoop `winutils.exe`:

To run Apache Spark on windows, you need Hadoop `winutils.exe`

Download them from:

<https://github.com/cdarlint/winutils/blob/master/hadoop-3.3.5/bin/winutils.exe>

<https://github.com/cdarlint/winutils/blob/master/hadoop-3.3.5/bin/hadoop.dll>

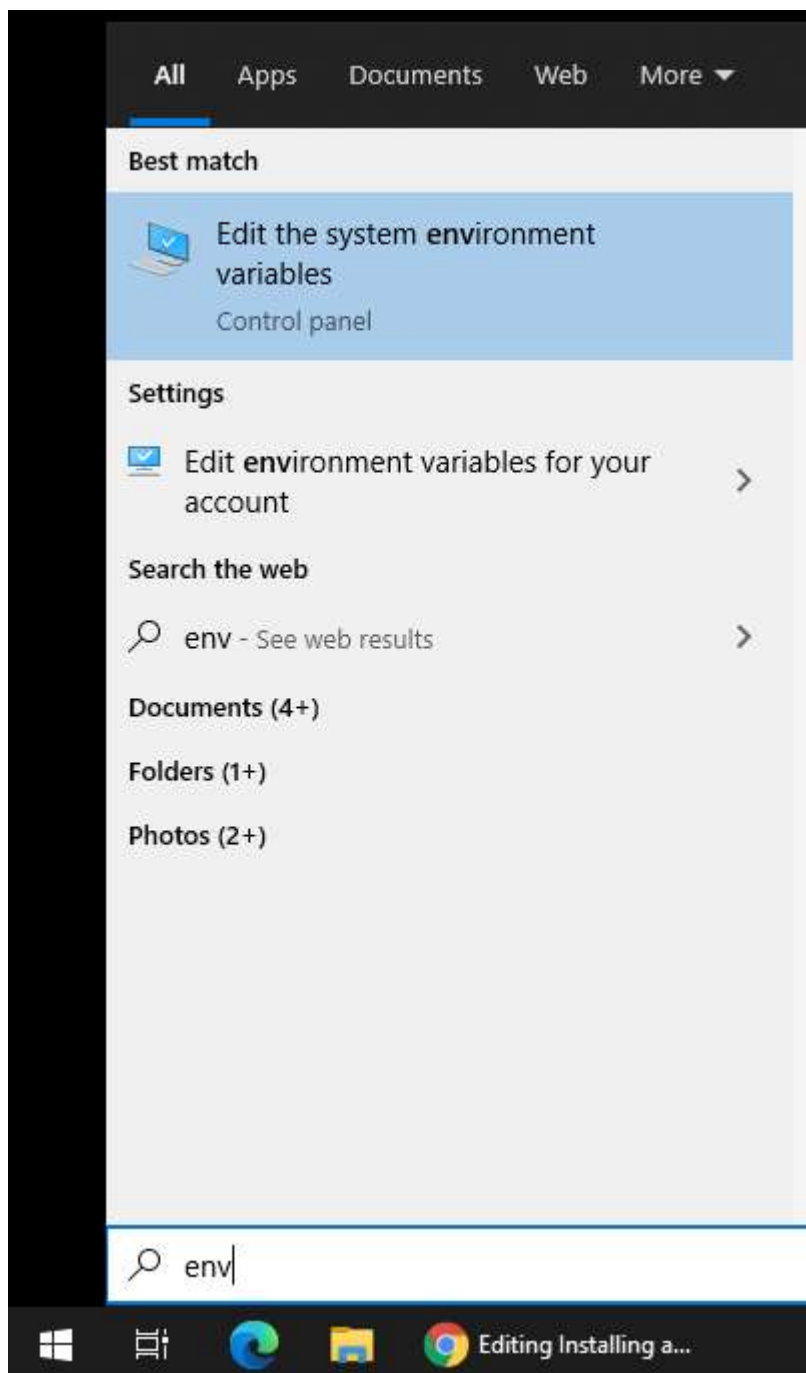
Make sure you download the `winutils.exe` and `hadoop.dll` files corresponding to the Spark and Hadoop version you are using.

Create a folder `C:\Hadoop` and a sub-folder named `bin` in it. Make sure you do not use spaces in the folder names. Now copy the files `winutils.exe` and `hadoop.dll` in the sub-folder `bin`.

Setting up Environment Variables:

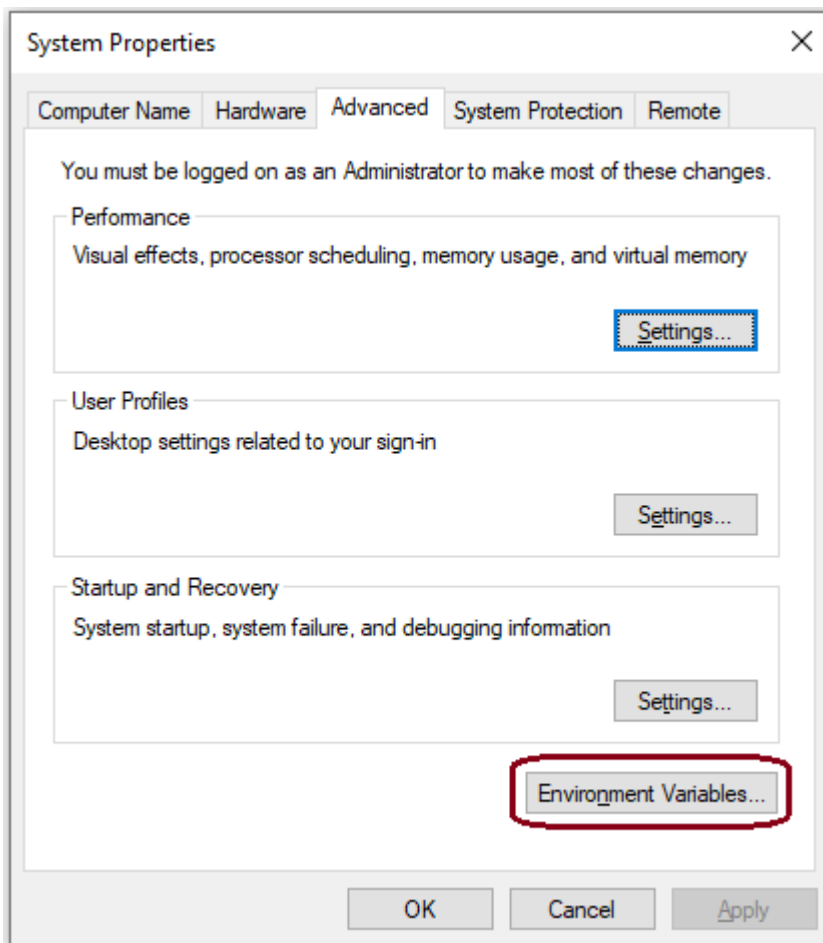
Environment variables need to be set so that PySpark can look up the correct folders for the required files.

In the Windows search box (textbox next to Windows icon in the status bar) type `env` to add and edit environment variables.



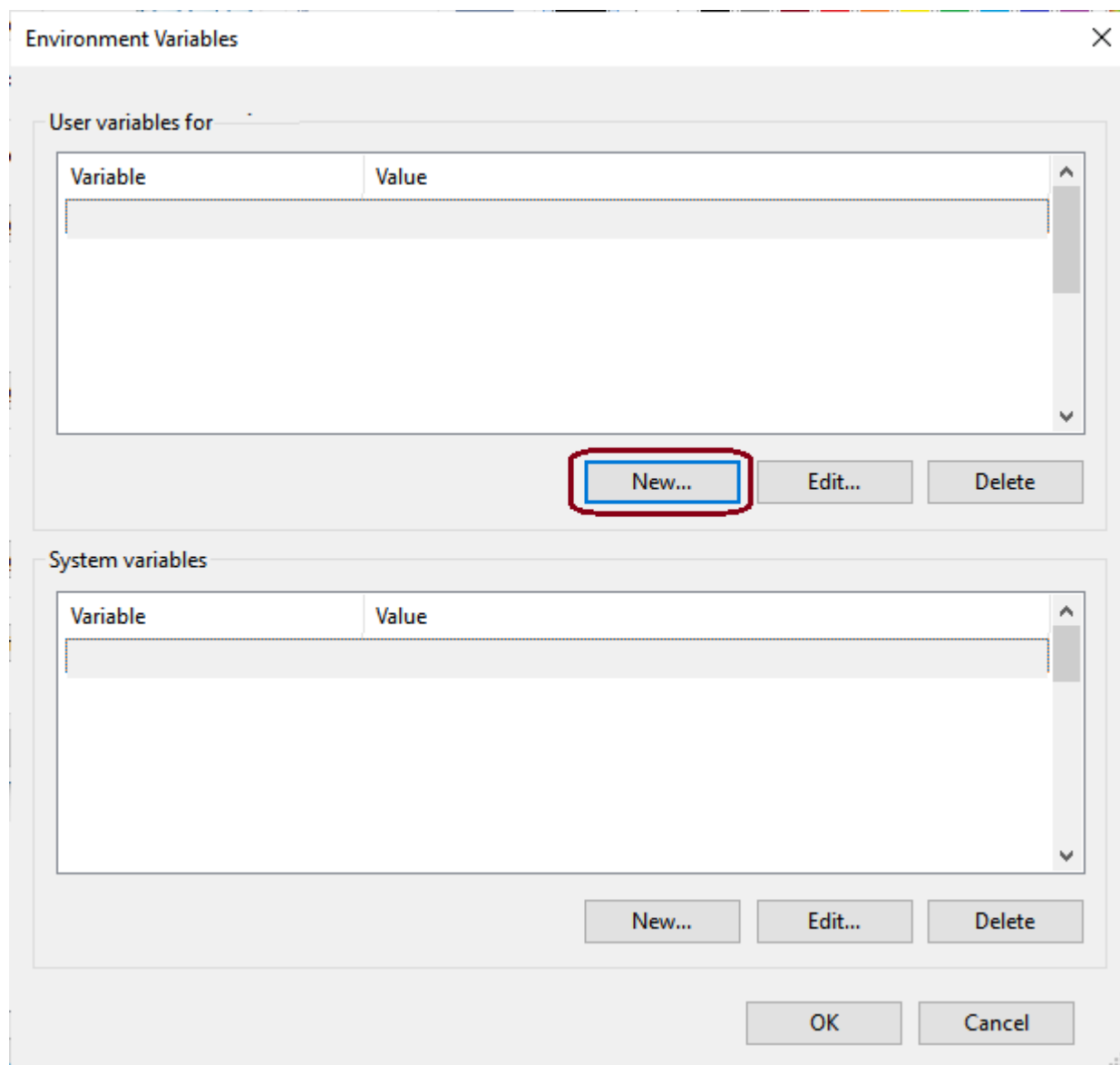
Click on the `Edit the system environment variables`

A window as shown below pops up. Click on `Environment Variables...` button.

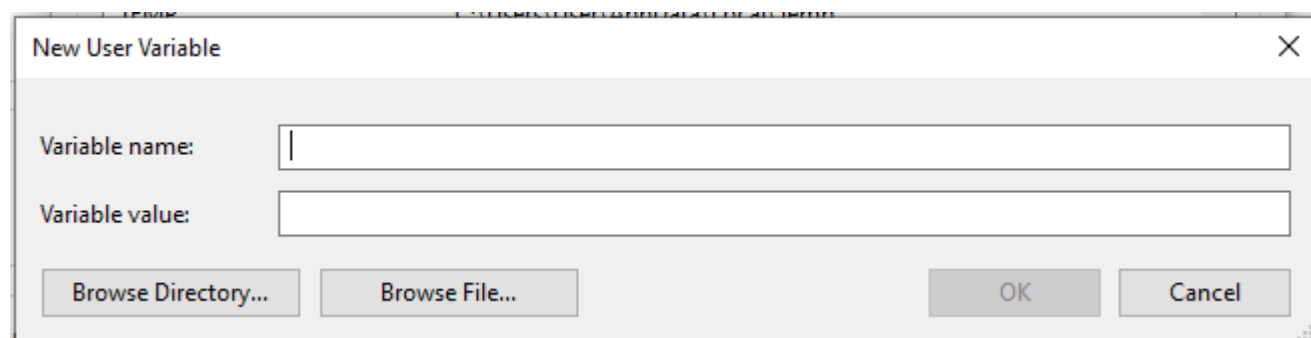


On the next screen (shown below) click on `New` button below `User variables for...` section.

On the next screen (shown below) click on **New** button below **User variables for...** section.



And add a new variable in the next screen as shown below.



Add each of the new variables listed below.

1. Variable name: `JAVA_HOME` (if not already present)

Variable value: <Name of the folder where Java is installed>

(Usually it will be `C:\Program Files\Java\jdk1.8.0_281` or `C:\Program Files (x86)\Java\jdk1.8.0_281`)

2. Variable name: `SPARK_HOME`

Variable value: `C:\Spark\spark-3.5.X-bin-hadoop3`

3. Variable name: `HADOOP_HOME`

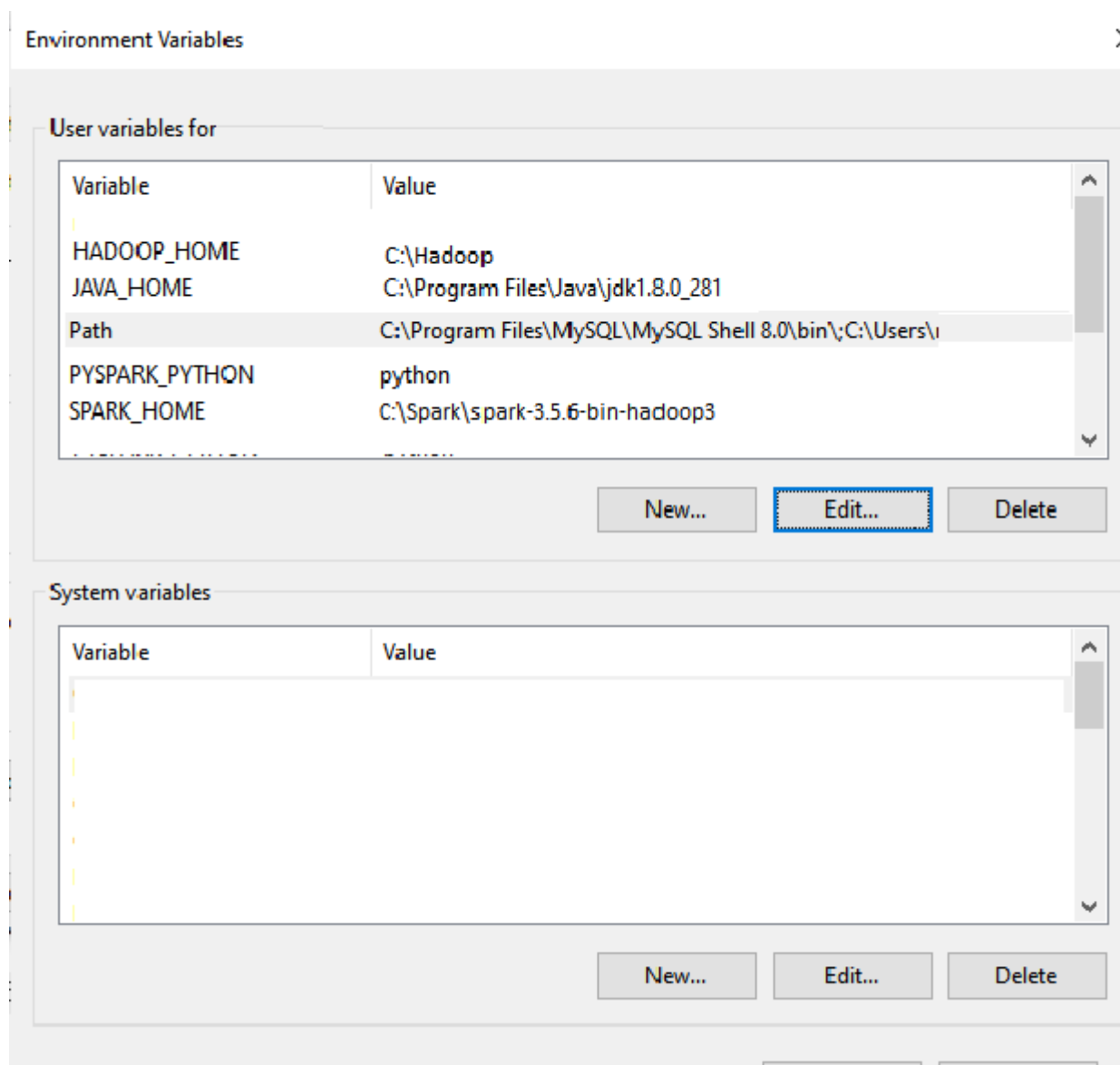
Variable value: `C:\Hadoop`

4. Variable name: `PYSPARK_PYTHON`

Variable value: `python`

5. Set the Path variable.

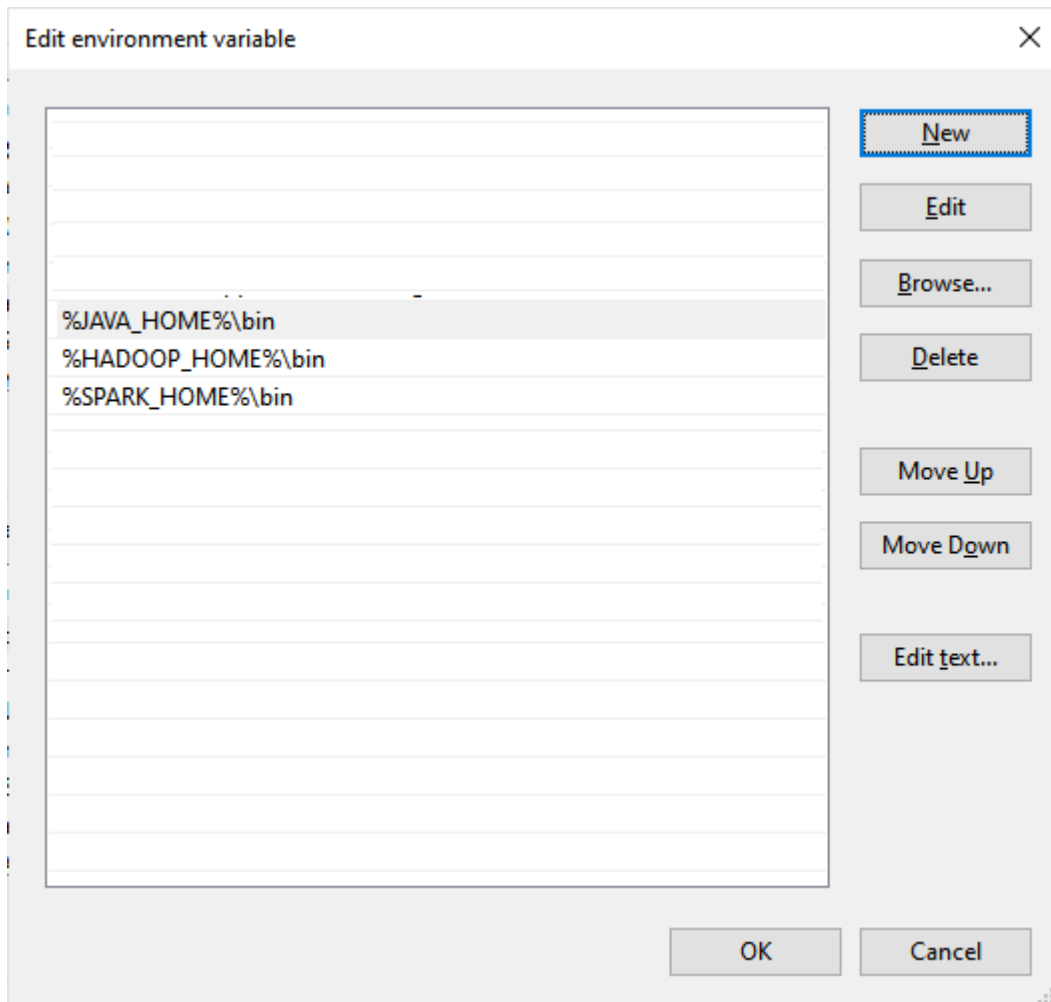
On Environment Variables window choose/highlight Path variable in User variables for... section and click on Edit button below this section.



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Note that the screen-shot above is given as an example; the folder names could be different; please use the names as given on your system.

In the next screen, click on the `New` button and add the variables as shown below.



1. %JAVA_HOME%\bin (if not present already)
2. %HADOOP_HOME%\bin
3. %SPARK_HOME%\bin

This completes the installation of Spark on the Windows.

You can run PySpark shell from the command prompt by giving the command `pyspark` and verify.